



METAGENOMICS WORKSHOP

*Metagenomic Next Generation Sequencing
for Pathogen Detection and Surveillance*

TABLE OF CONTENTS

Overview of Workshop	2
Abbreviation Table	4
Molecular Biology Essentials: The Central Dogma	6
mNGS Workflow for Pathogen Detection	7
Study Design	7
Nucleic Acid Extraction	11
Library Preparation	12
Illumina Sequencing	13
Tips for Accurate Pipetting	14
Module 1: Hands-on RNA Library Preparation and Sequencing	15
WL- Protocol 1: SPRI Bead Practice	17
Workshop Sample Set List	21
WL- Protocol 2: Illumina RNA Library Preparation (Watchmaker reagents)	22
WL- Protocol 3: Illumina RNA Library QC & Pooling	31
WL- Protocol 4: Sequencer Loading	37
Activity: Basespace Run Review	39
Module 2: Analysis and Interpretation of Illumina RNA mNGS data	41
CZ ID Training: Reference Tables	42
CZ ID Training: Worksheets	47
Appendix:	60
Appendix: DNase I treatment protocol	60
Appendix: Secondary Amplification protocol	61
Appendix: TruSeq-compatible mini adapter info sheet	63

OVERVIEW OF MNGS WORKFLOW

Metagenomic Next-Generation Sequencing (mNGS) is an essential tool for pathogen detection, discovery, and surveillance. This exercise will have two modules: laboratory theory and hands-on sessions, followed by no-code data analysis and interpretation. During the first week, participants will generate metagenomic libraries in the lab, gaining skills in Illumina library preparation, troubleshooting, and sequencing. Subsequently, the second week will center on the analysis and interpretation of metagenomic sequencing data using the [CZ ID platform](#).

Hands-on RNA Library Preparation and Sequencing

Day	Main Objectives	Lesson plan
Monday	• Introduce mNGS concepts • Review study design • Walkthrough sample collection, handling, storage, and hands-on RNA extraction	• Lecture: "Introduction to mNGS" • Lecture: "Study design" • Hands-on: Walkthrough of typical workflows + feedback • Hands-on: RNA extraction + QC
Tuesday	• Understand the lab process to prepare Illumina libraries • Practice cleaning, size selecting, and quantifying DNA • Library Prep Day 1: synthesize dsDNA for mNGS library preparation	• Lecture: "From sample to sequences" • Hands-on: SPRI bead clean-up • Hands-on: Follow protocol for RNA fragmentation and dsDNA synthesis
Wednesday	• Library Prep Day 2: Ligate adapters to end repaired DNA fragments synthesized on Library Prep Day 1 • Understand quality control of Illumina sequencing libraries • Perform quality control on the libraries prepared during the training	• Hands-on: Follow protocol for end repair, adapter ligation and barcoding PCR • Hands-on: Follow protocol for library clean-up, size selection • Lecture: "Illumina library QC & pooling" • Hands-on: Library quality control and pooling
Thursday	• Understand the principles underlying Illumina sequencing • Load and sequence a library pool on an iSeq and/or NextSeq	• Lecture: "Illumina sequencing theory" • Hands-on: iSeq loading

Analysis and Interpretation of Sequencing Data Utilizing CZ ID

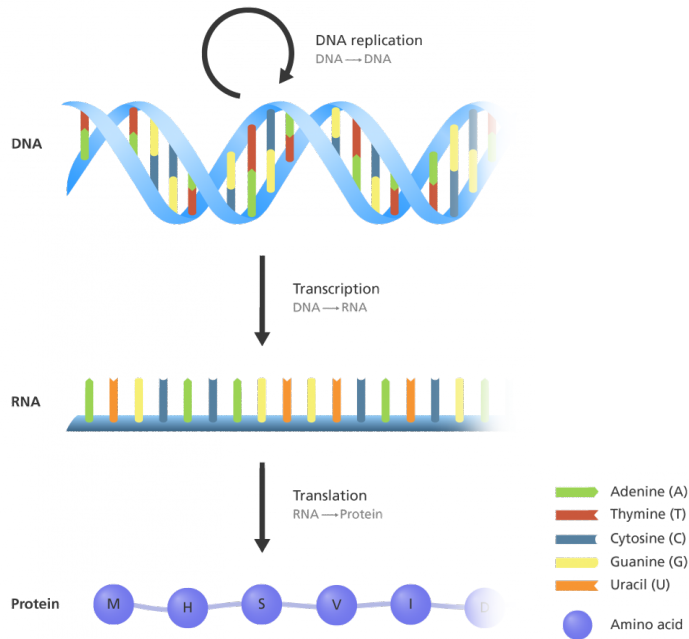
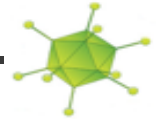
Day	Main Objectives	Lesson plan
Sunday	- Evaluate sequencing run quality - Hands-on practice identifying sequences using BLAST - Introduction to the CZ ID platform	- Basespace run review lecture and activity - Lecture: "Bioinformatics basics" - Activity: BLAST - Lecture: "Intro to CZ ID"
Monday	- Understand the CZ ID mNGS pipeline and how to interpret results for a single sample in CZ ID - Hands-on practice interpreting mNGS results for single samples in CZ ID - Discuss contamination issues and how we can address it in CZ ID - Understand consensus genomes and assembly metrics to assess genome quality - Hands-on practice evaluating genome quality and completeness	- Lecture: "CZ ID mNGS analysis" - Activity: Look at data! (real data from week 1 sequencing) - Lecture: "What about contamination?" - Lecture: "Assembling viral genomes in CZ ID" - Activity: Evaluate genome assembly
Tuesday	- Learn how to use heatmaps in CZ ID to visualize taxa detected across samples - Hands-on practice interpreting heatmaps - Become familiar with other sequencing workflows - Putting it all together	- Lecture: "Heatmap analysis" - Activity: Heatmap analysis activity - Lecture: "mNGS enrichment strategies" - Unstructured time for review and summarizing

ABBREVIATION TABLE

Abbreviation	Description
16s rRNA	Component of the structural ribosomal RNA (rRNA) for the small subunit (SSU) of prokaryotic cytoplasmic ribosomes. The S stands for Svedberg unit, which measures sedimentation coefficients
18s rRNA	Component of the structural ribosomal RNA (rRNA) for the small subunit (SSU) of eukaryotic cytoplasmic ribosomes. The S stands for Svedberg unit, which measures sedimentation coefficients
28S rRNA	Structural ribosomal RNA (rRNA) for the large subunit (LSU) of eukaryotic cytoplasmic ribosomes. The S stands for Svedberg unit, which measures sedimentation coefficients
BLAST	Basic Local Alignment Search Tool and refers to a suite of programs used to generate alignments between a nucleotide or protein sequence, referred to as a “query” and nucleotide or protein sequences within a database, referred to as “subject” sequences
bp	Base Pair: a unit of measurement used to designate the length (in number of nucleic acid bases) of DNA or RNA
BSL1	As the lowest of the four, biosafety level 1 applies to laboratory settings in which personnel work with low-risk microbes that pose little to no threat of infection in healthy adults. An example of a microbe that is typically worked with at a BSL-1 is a nonpathogenic strain of <i>E. coli</i> .
BSL2	This biosafety level covers laboratories that work with agents associated with human diseases (i.e. pathogenic or infectious organisms) that pose a moderate health hazard
Ct	Cycle threshold: Number of PCR cycles required for a fluorescent signal to exceed background level (see 2.1.2. Sample Storage section)
DNA	Deoxyribonucleic acid
dsDNA	Double-Stranded Deoxyribonucleic acid
EDTA	Ethylenediaminetetraacetic acid: a metal ion chelator, often added to reactions in molecular biology to reduce the activity of unwanted enzymes and stabilize nucleic acids
ERCC controls	A common set of synthetic RNA molecules typically used as a spike-in positive control to sequencing experiments to measure variability in/between experiments. It was developed by the External RNA Controls Consortium, where the name comes from
gDNA	Genomic deoxyribonucleic acid is chromosomal DNA, in contrast to extrachromosomal DNAs like plasmids or organelle DNA
HS	High Sensitivity

Abbreviation	Description
HT1	Illumina proprietary formula of the hybridization buffer used to dilute denatured libraries and dilute the PhiX control library prior to sequencing
Kb	Kilobase, a unit of measurement used to help designate the length of DNA or RNA. One kilobase is equal to 1000 bases
mNGS	Metagenomic Next-Generation Sequencing
NGS	Next-Generation Sequencing
NP swab	Nasopharyngeal swab: A sample taken by inserting a swab into the nostril parallel to the chin, until resistance is felt. The specimen is taken from the nasopharynx, which is the upper part of the throat behind the nose.
OP swab	Oropharyngeal swab, A sample taken by inserting a swab through the oral cavity. The specimen is collected from the rear wall of the oropharynx.
PCR	Polymerase chain reaction: a technique in molecular biology to copy sequence-specific regions of DNA
PEG	Polyethylene glycol is a polyether compound. PEG is also known as polyethylene oxide or polyoxyethylene, depending on its molecular weight. The structure of PEG is commonly expressed as $H-n-OH$
QC	Quality control
RNA	Ribonucleic acid
rRNA	Ribosomal RNA
RT	Room Temperature: conventionally around 20°C
SPRI Beads	Solid Phase Reverse Immobilization Beads: used in molecular biology to purify and/or size select nucleic acids
TBE	A buffer solution containing a mixture of Tris (tris-hydroxymethyl aminomethane) base, boric acid and EDTA
TNA	Total Nucleic Acid: a combination of DNA and RNA
xNA	x-Nucleic Acid: DNA or RNA

MOLECULAR BIOLOGY ESSENTIALS: THE CENTRAL DOGMA



Borrowing from Viruses

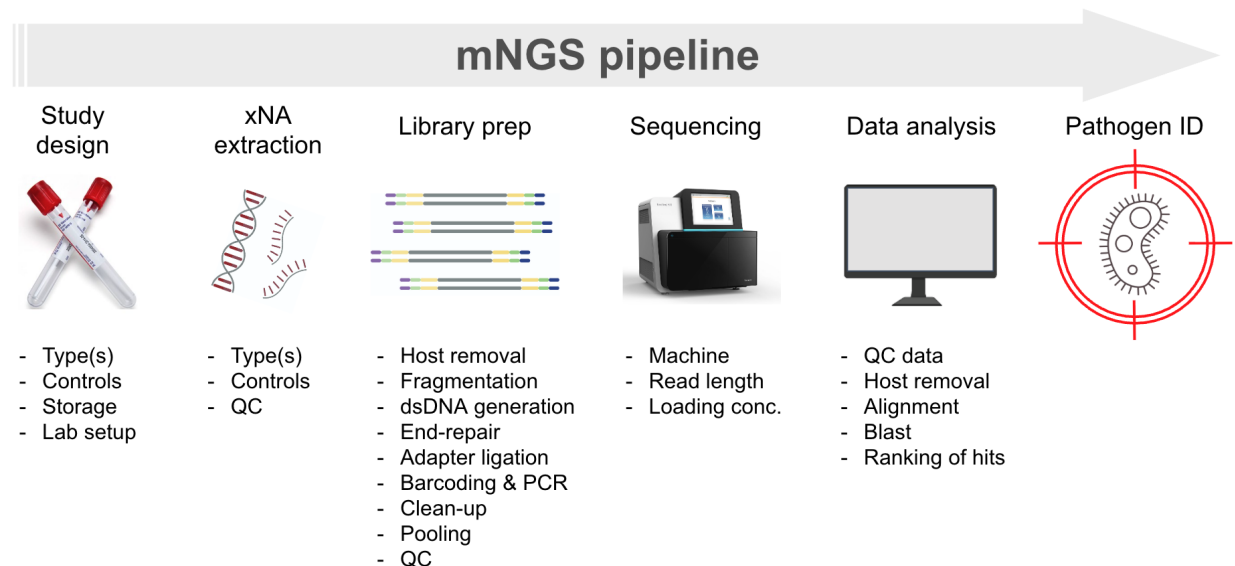
In the 70s it was discovered that RNA viruses contain enzymes that use viral RNA as a template for the synthesis of DNA, thereby reversing the direction of transcription. Since then, we have incorporated reverse transcription and the synthesis of cDNA as a molecular biology technique, to study gene expression and RNA genomes.



	Viruses	Bacteria	Eukaryotic microbes (yeast, amoebas, etc.)
Genome composition	ssRNA, ssDNA, dsDNA, or dsRNA	dsDNA	dsDNA
Ribosomal RNA	None, uses host ribosomes	16S rRNA	18S rRNA
Genome size	2 Kb - 1 Mb	2 Mb - 8 Mb	200 Kb - 100,000 Mb

Metagenomic Next-Generation Sequencing (mNGS) is a powerful tool to sequence all of the genetic material in a sample. We will focus on metagenomic Next Generation Sequencing (mNGS) of RNA to identify potential pathogens in nasopharyngeal swabs.

MNGS WORKFLOW FOR PATHOGEN DETECTION

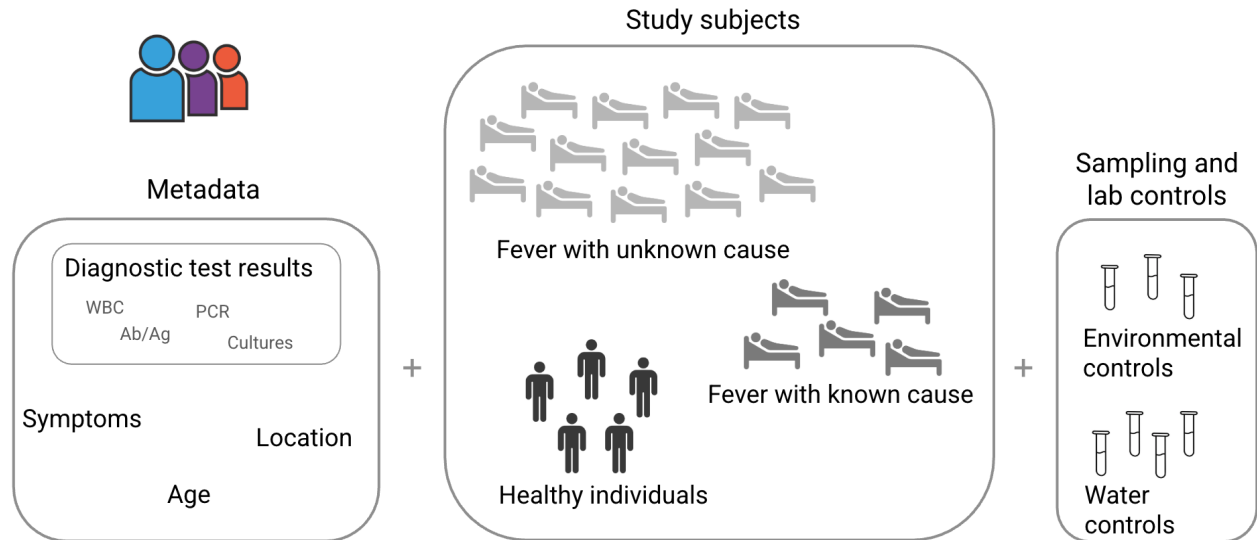


Study Design

Study Subjects, Controls and Metadata

After we have a defined objective and research question, we can proceed with collecting samples. There are many considerations at this stage and the first is determining the study subjects and proper controls; Positive controls are diseased individuals with known etiologies, Negative controls are healthy individuals and Sampling controls are mock samples. In mock samples, the sampling medium is exposed to environmental contaminants of sampling collection sites. Sampling controls will provide a background for environmental contamination. During library preparation, additional water controls will be added to survey environmental contamination and cross-sample contamination. The samples should have as much metadata associated with them as possible. Integrating the metadata with sequencing results will help strengthen conclusions/interpretation.

Study Subjects, Controls and Metadata (continued)

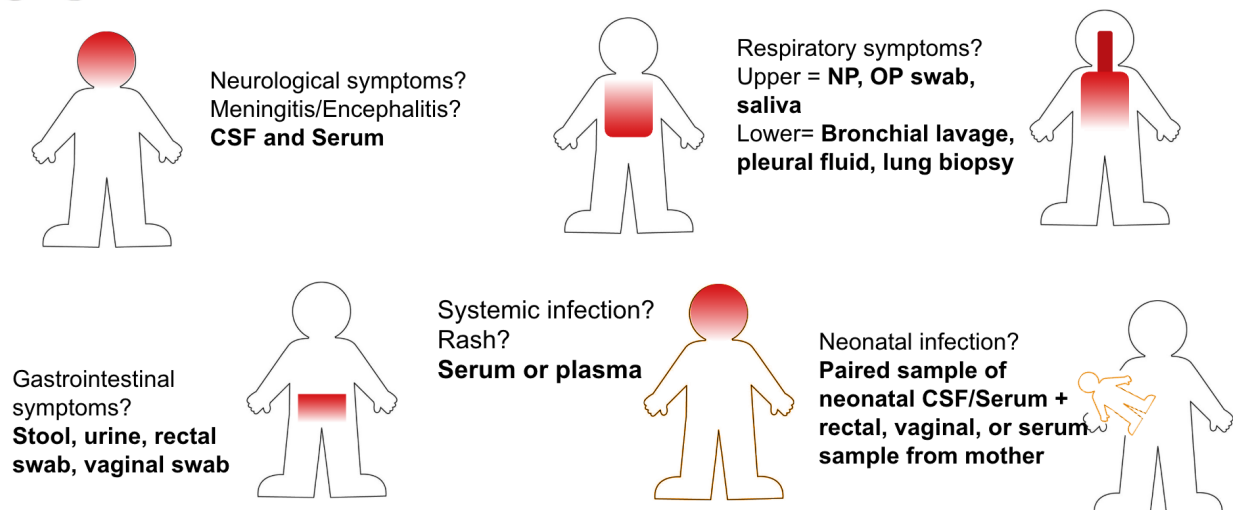


Sample Types

The type of sample collected from the patient and the timing of collection since symptom onset are critical to study design. These should be selected considering the study question, patient symptomatology, and timing of symptom onset.



Collected sample type depends on the study question



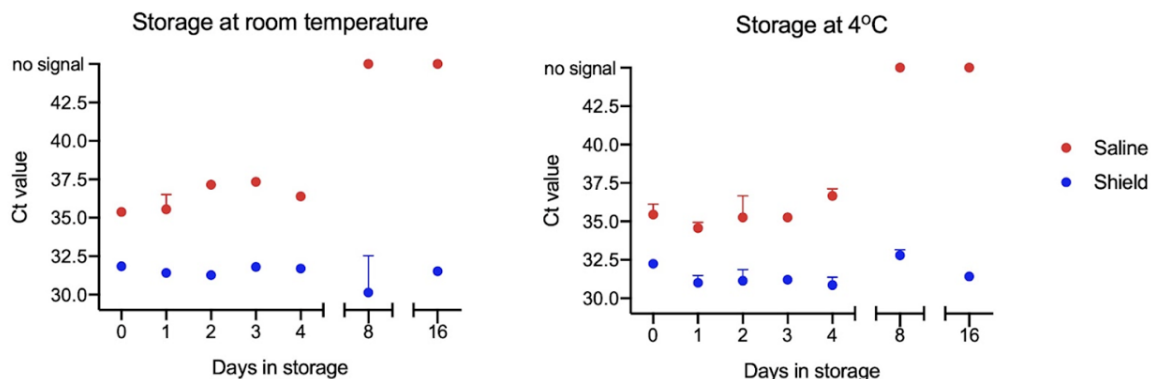
Sample Handling and Storage

DNA and RNA are prone to degradation because of various factors such as temperature, UV radiation, enzymes (DNases and RNases from the environment), and pH. RNA is extremely delicate and can easily degrade if not kept at low temperatures and in nucleic acid stabilizers. Good quality nucleic acid is the first step to sequencing success.

To preserve sample integrity:

- Collect and store in nucleic acid stabilizers (Zymo DNA/RNA Shield*)
**DNA/RNA Shield is incompatible with bleach and dangerous if mixed*
- Collect samples using sterile techniques. Keep at 4°C or less during transportation
- Keep at -80°C for long term storage
- Minimize the number of freeze-thaw cycles

Bonus Info:



In vitro transcribed RNA fragments preserve better in Shield than in Saline when stored at room temperature or at 4°C, as assessed by qPCR.

What does Ct mean?

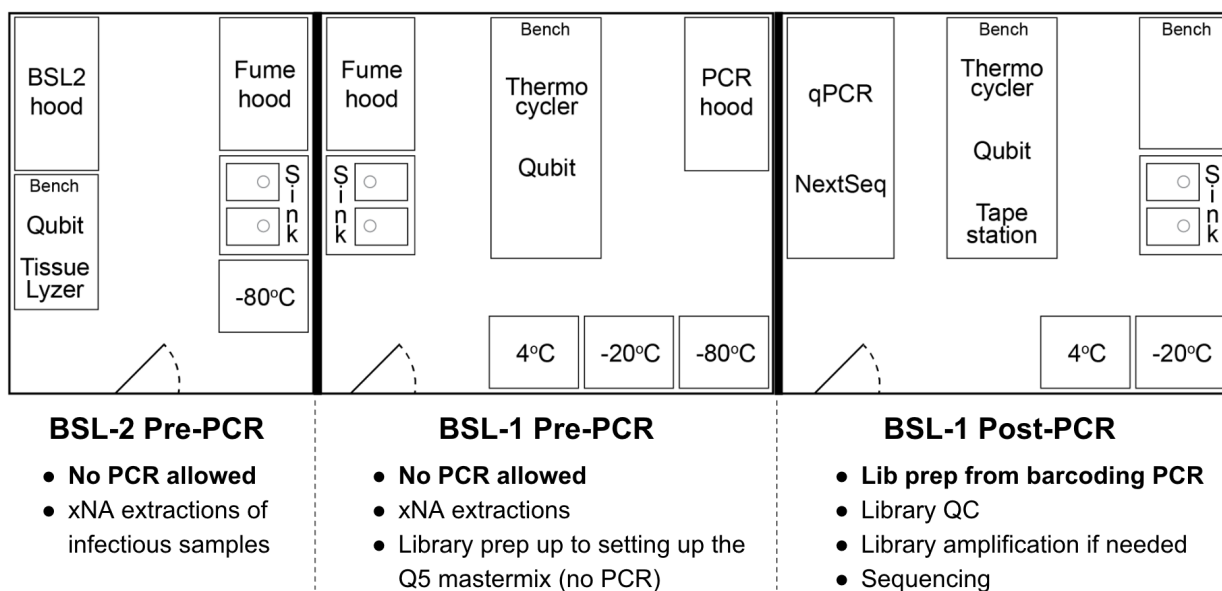
In a real time PCR assay, a positive reaction is detected by accumulation of a fluorescent signal. The Ct (cycle threshold) is defined as the number of cycles required for the fluorescent signal to cross the threshold (i.e. it exceeds background level). Ct levels are inversely proportional to the amount of target nucleic acid in the sample (i.e. the lower the Ct level the greater the amount of target nucleic acid in the sample). Cts < 29 are strong positive reactions indicative of abundant target nucleic acid in the sample, Cts of 30-37 are positive reactions indicative of moderate amounts of target nucleic acid, Cts of 38-40 are weak reactions indicative of minimal amounts of target nucleic acid which could represent an infection state or environmental contamination.

* RNases are your enemy in the beginning of this workflow because they degrade RNA, which is what you are looking for!

Laboratory Setup

A good laboratory setup can help reduce contamination and reduce RNases by keeping pre-PCR and post-PCR spaces separate. Ideally, a laboratory will have dedicated areas for different procedures. The areas can be classified by:

- The **biosafety level** of the samples: Prior to inactivation, infectious agents must be handled at higher biosafety levels. Non-pathogenic microbes or inactivated samples can be handled at lower biosafety levels. Inactivation of biological samples is the process by which potential pathogens are rendered non-infectious. DNA/RNA Shield is an inactivating reagent.
- **Pre- and post-PCR**: PCR products are highly enriched DNA molecules that can cause contamination to other samples and future experiments, especially those with low starting material. Thus, pre-PCR areas should be free of PCR products.
- **PCR hoods** are enclosed workspaces that provide a space for handling xNA (DNA and RNA) in order to avoid environmental and cross contamination.



Additional Precautions: We want to reduce cross-sample and cross-experiment contamination and reduce RNases to maintain the integrity of our RNA samples. Here are a few recommended precautions:

- Decontaminate surfaces and gloves with RNaseZap and Ethanol (70%).
- Aliquot reagents in pre-PCR clean spaces, ideally with dedicated pipettes.
- Avoid talking while handling open tubes and while aliquoting reagents.
- Make sure to always place samples and reagents on ice to keep them cold.
- Use filter tips and always switch pipette tips between samples.

Nucleic Acid Extraction



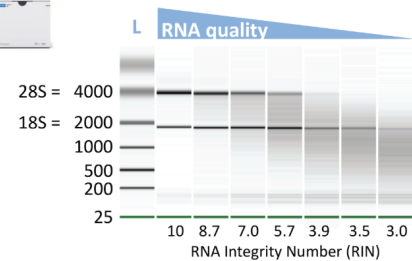
- Before extraction:
 - Spin down to remove host cells (extract supernatant)
- Choose type:
 - DNA:
 - Detect bacteria, parasites, fungi, DNA viruses
 - RNA:
 - All the above + RNA viruses (!)
 - Requires optimal sample quality, degrades very fast
 - Keep cold at all times!
 - TNA (total nucleic acid):
 - DNA + RNA
 - Requires DNase I step for RNA-mNGS
- Column vs bead-based extraction

Key points:

- ✓ Minimize cross-contamination
- ✓ Preserve sample quality



- Quality control:
 - Quantity (ng/μl):
 - Qubit: RNA or DNA High Sensitivity (HS) Assay Kit
 - Quality (length):
 - Capillary electrophoresis
 - TapeStation (Agilent)
 - High Sensitivity DNA or RNA ScreenTape
 - Gel electrophoresis
 - Regular agarose gel for DNA
 - TBE-urea gel for RNA

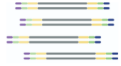


Key points:

- ✓ NA quality is crucial
- ✓ Knowing NA quality can help with troubleshooting and interpretation

Library Preparation

Steps in Library preparation:

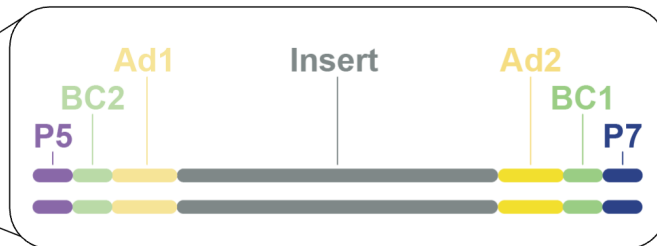


1. Host depletion, RNA fragmentation + priming
2. First strand cDNA synthesis
3. Second strand cDNA synthesis
4. End Prep
5. Adapter Ligation
6. Barcoding PCR

Sequencing library = collection of fragments ready for sequencing

1. RNA is copied to cDNA
2. Adapter sequences, barcode sequences and flow cell binding sites are added to allow the sequencing process to occur

Sample library

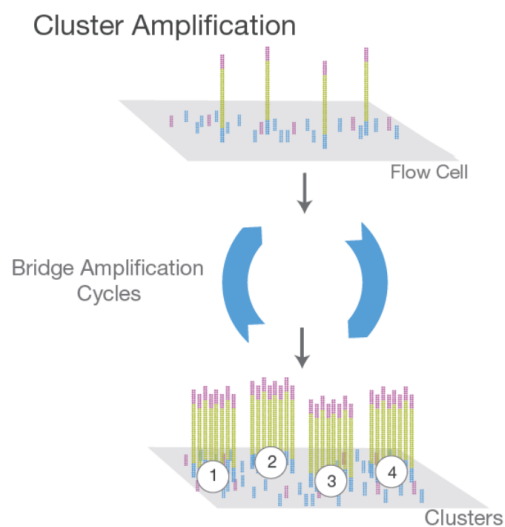


- | | |
|----------------|---|
| Insert | = DNA fragment to be sequenced |
| Ad1/Ad2 | = Adapters that allow initiation of sequencing |
| BC1/BC2 | = Barcodes specific for each sample (one for each strand) |
| P5/P7 | = Flow cell binding sites |

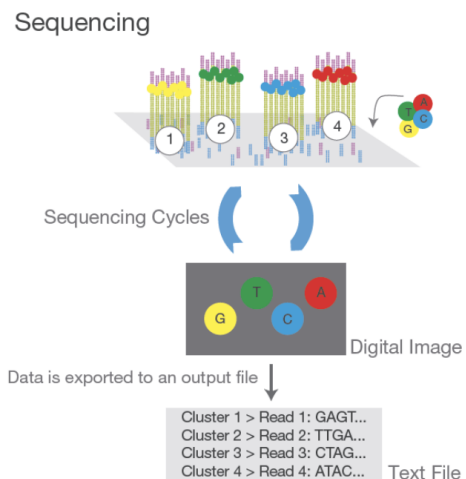
Illumina Sequencing

<https://www.illumina.com/science/technology/next-generation-sequencing/sequencing-technology.html>

Illumina platforms use Sequencing by Synthesis (SBS) technology. In this method, a copy of the DNA fragment is made and each base is “read” by imaging the fluorescent signal of each nucleotide as it is added to the template. Before SBS begins, library molecules bind the flow cell and are clonally amplified into “clusters” in order to amplify the signal that will come from the SBS process. The raw data collected is images from each cycle of sequencing.



Library is loaded into a flow cell and the fragments hybridize to the flow cell surface. Each bound fragment is amplified into a clonal cluster through bridge amplification.



Sequencing reagents, including fluorescently labeled nucleotides, are added and the first base is incorporated. The flow cell is imaged and the emission from each cluster is recorded. The emission wavelength and intensity are used to identify the base. This cycle is repeated “n” times to create a read length of “n” bases.

Tips for Accurate Pipetting

1. Pipettes have particular ranges of volumes in which they are accurate. For a given volume it is important to choose a pipette in which the desired volume falls within the pipette's range. *(The closer to the middle of the range the better. For instance, to pipette 100 μ l, you would use a P200 as opposed to a P1000.)*
2. Make sure to press the tip gently but firmly onto the pipette, because a loose tip will not draw the proper volume of liquid. Avoid slamming the pipette into the tips.
3. Slowly push down to the first stop of the plunger before submerging the tip into the liquid.
4. Submerge the tip 3 to 4 mm into the liquid. The more you submerge, the more liquid may stick to the outside of the tip. *(Make sure the tip remains submerged as liquid is brought into the pipette.)*
5. Slowly release the plunger of the pipette after the tip has been submerged. The speed at which you release will depend on the type of liquid - more viscous liquids require slower aspiration.
6. Hold the pipette vertically when aspirating, moving, and dispensing liquid.
7. Hold the tube you are pipetting at an angle (~45 degrees) and pipette either into the liquid in the tube or on the side of the tube near the bottom.
8. Eject the used tip into an adequate disposal container.
9. When making master mixes, make enough for your samples plus at minimum 10% extra to protect against pipetting variation. For example, if the master mix is for 10 samples, calculate enough volume for 11 samples ($10 \times 1.1 = 11$).
10. When instructed to "mix by pipetting", set your pipette to ~80% of the total volume in the tube and mix by pipetting up and down at least 10x. Visually inspect the mixture to confirm it is homogenous. Pro tip: when mixing, draw up from the bottom of the liquid, dispense near the top of the liquid.

MODULE 1: IN THE LAB

Hands-on RNA Library Preparation and Sequencing

MODULE 1: HANDS-ON RNA LIBRARY PREPARATION AND SEQUENCING

Getting Started

- Before starting and after completing your work, always wipe the workspace with RNaseZap/RNaseAway and 70% EtOH
- Tie back long hair
- Wear a clean lab coat
- Change out gloves often
- Organize your workspace in a way that avoids hovering over clean tips/tubes with a used pipette
- Only keep necessary materials out for the protocol you are performing
- Clean the workspace at the end of the day with 70% EtOH and dispose of trash appropriately

For more information on RNAses, we recommend visiting:

<https://www.thermofisher.com/us/en/home/references/ambion-tech-support/nuclease-enzymes/tech-notes/how-to-maintain-an-rnase-free-lab.ht>

List of Protocols

Below is the list of protocols that will be used during the wet lab module:

- Protocol 1: SPRI Bead Practice
- Protocol 2: Illumina RNA Library Preparation
- Protocol 3: Illumina RNA Library QC & Pooling
- Protocol 4: Sequencer Loading

WL- Protocol 1: SPRI Bead Practice

Overview

SPRI (Solid Phase Reverse Immobilization) beads are useful for purifying and size-selecting nucleic acids such as PCR amplicons, NGS libraries, RNA/DNA, and gDNA. As demonstrated in the figure below, nucleic acids bind to the SPRI beads, are washed, then eluted from the beads before collection. Adjusting the ratio of SPRI beads to sample volume dictates selection for nucleic acids by base pair size.

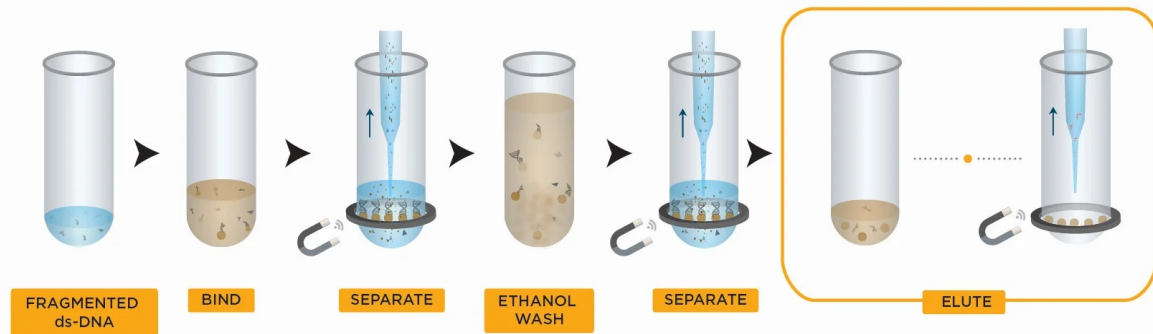


Image from: <https://www.beckman.com/reagents/genomic/cleanup-and-size-selection/size-selection>.

It is crucial that all reagents and materials are sterile and RNase-free.

Reagents and Materials

Reagent	Vendor	Catalog No.
SPRISelect Bead-Based Reagent	Beckman Coulter	B23319
Molecular-grade ethanol	Various	-
Nuclease-free water	Ambion	AM9937
50bp ladder	Invitrogen	10416-14
Gel Electrophoresis system	-	-
Qubit system + dsDNA HS reagents	Thermofisher	Q32854

SPRI Clean Protocol

Use [SPRI Select](#), [Ampure XP](#), or [Illumina Purification](#) beads. They work equivalently and only differ in storage conditions. This protocol is an adaptation of the manufacturer instructions.

Prepare fresh 80% EtOH - 500ul per sample (this includes enough overage volume).

Based on the desired SPRI ratio and sample volume, calculate the volume of SPRI beads to add. Also decide on a desired final volume after clean-up.

Sample volume:	
SPRI bead ratio:	
SPRI to add (per rxn):	
Elution volume (per rxn):	
Aspiration volume (per rxn):	

1. Add SPRI beads to sample and mix thoroughly by pipetting up and down ten times.
2. Incubate for 3-5 mins at room temperature. Then pulse spin the tubes very briefly (1-2s), but ensure beads are still fully suspended.
3. Place samples on magnetic rack and incubate for ~5 mins on the rack.
4. Remove supernatant.
5. Add 200µL of 80% EtOH to samples while on the magnetic rack. Do not disturb beads. Incubate at room temperature for 30s then remove the supernatant.
6. Repeat EtOH wash step for a total of 2 washes.
7. Remove any excess EtOH with a p10 or p20 pipette.
8. Air dry the beads for up to 3-5 mins while on the magnetic rack.
Watch beads carefully throughout this incubation to avoid over-drying; smaller bead pellets dry faster. Trust your eyes more than the timer.
9. Remove tube from magnetic rack. Add elution volume of nuclease free water.
10. Vortex or pipette to mix well. Incubate for a *minimum* of 2 mins at room temperature off the magnetic rack.
11. Pulse spin tubes and place on magnetic rack until solution is clear ~ 5 mins.
12. Remove desired aspiration volume of the supernatant and transfer to a clean nuclease free PCR tube

SPRI Bead Clean-up Practice

We will practice SPRI bead cleaning by performing size selections at different bead:sample ratios on a 50bp or 100bp DNA ladder at a **1:50 dilution**.

1. Create a 1:50 dilution of DNA ladder in a labeled 1.5mL tube.
2. Keep an aliquot of the 1:50 dilution for HS dsDNA Qubit quantification and for a capillary gel electrophoresis run.
3. Prepare sufficient 80% EtOH
4. Prepare labeled PCR strip tubes, then perform a SPRI clean-up using the protocol in the previous page for the ratios and volumes below

Well	1	2
Sample	20 µL ladder	20 µL ladder
SPRI Ratio	0.5x	0.8x
SPRI bead volume	10 µL	16 µL

Check DNA Elution Using dsDNA Qubit

1. Quantify the initial 1:50 diluted 50bp ladder and all the SPRI eluates by HS dsDNA Qubit to assess yield.
2. Follow the instructions provided by the manufacturer. Each HS dsDNA Qubit assay requires at least 1 µL of sample.

Input ladder concentration =

Input ladder mass =

Sample	1	2
Concentration (ng/µL)		
Final Mass (ng)		
Efficiency (%)		

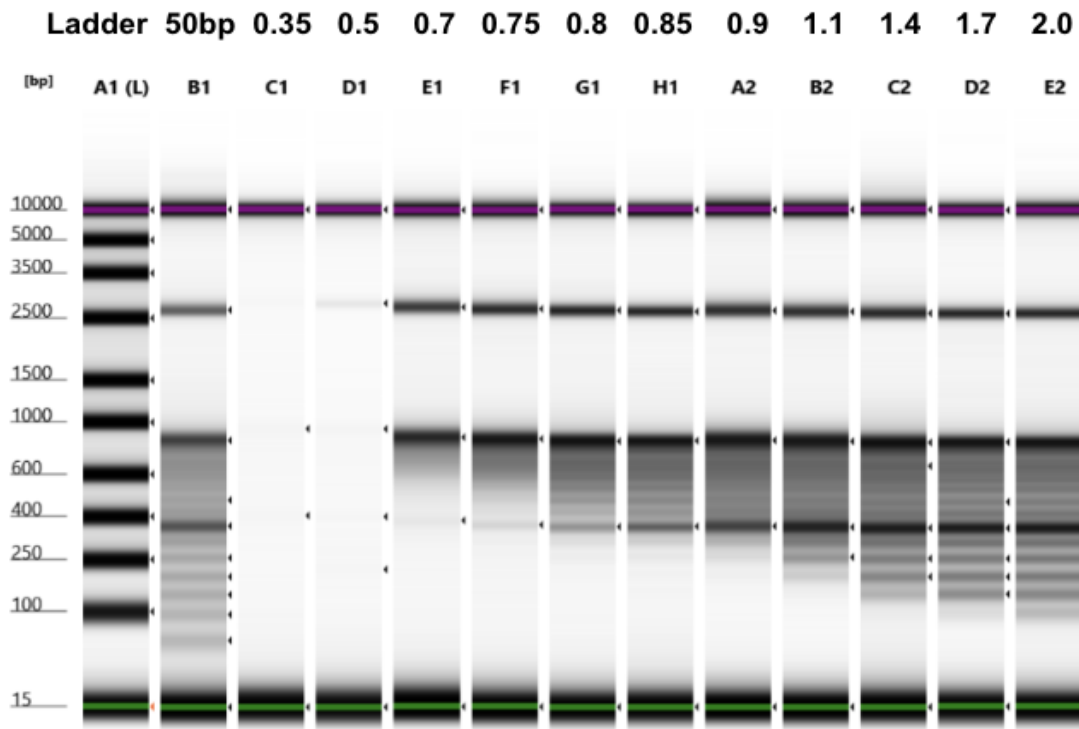
*Final Mass = Elution volume * Elution concentration*

*% Yield = (Final Mass / Input Mass) * 100*

Check DNA Elution Using Gel Electrophoresis

Samples may be run on a 1.8% agarose gel which is more cost effective or on a capillary electrophoresis platform such as a Tapestation.

Typical results:



Results of different SPRI bead clean ratios on a 50bp ladder, run on a Tapestation D5000

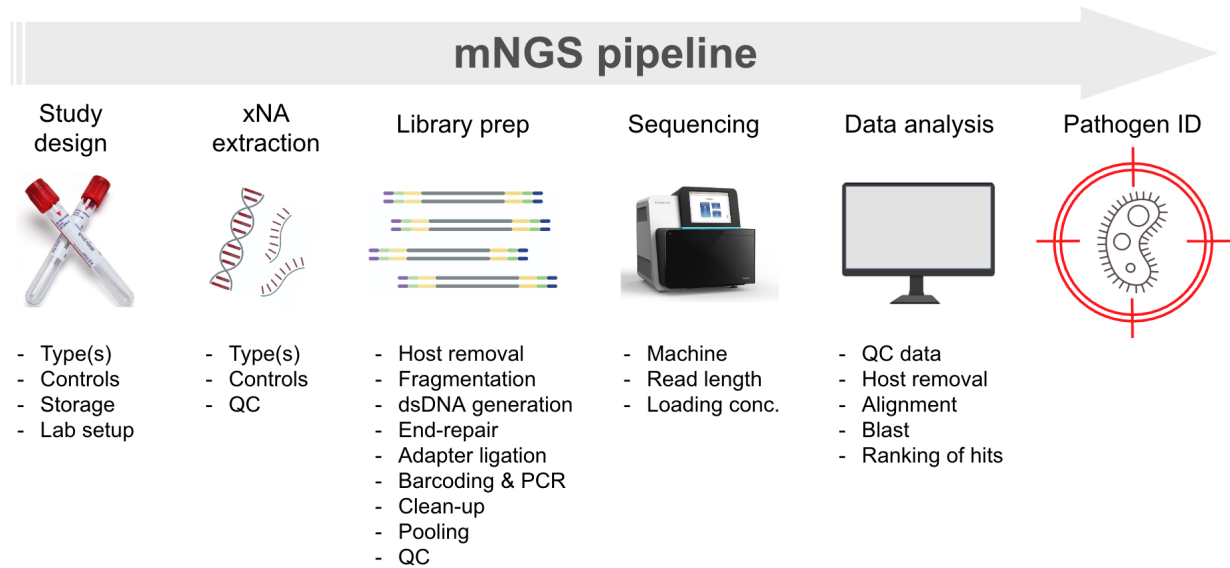
Sample list - RNA mNGS pilot

Workshop ID	Sample category	Sample type	Input RNA concentration (ng/ul)	Index / Barcode	Library concentration (ng/ul)
P_1					
P_2					
P_3					
P_4					
P_5					
P_6					
P_7					
P_8					
G_1					
G_2					
G_3					
G_4					
G_5					
G_6					
G_7					
G_8					

Sample set selection per batch

- Each batch must include at minimum 1 negative water control included from extraction through library prep
- It is recommended to include 1 positive control sample that has already tested positive by PCR or sequencing

WL- Protocol 2: Illumina RNA Library Preparation



RNA mNGS Library Preparation Workflow

The steps below are completed after study design and sample selection and collection.

0. RNA extraction and DNase I treatment
1. FastSelect + RNA Fragmentation + /Priming
2. First strand cDNA synthesis
3. Second strand cDNA synthesis and A-Tailing
4. Adapter Ligation
5. SPRI clean-up with size selection
6. Barcoding PCR
7. SPRI clean-up with size selection
8. Library QC
9. Sequencing
10. Sequencing run QC
11. Data QC and analysis

Considerations for library preparation

- All steps should be conducted in a clean RNase-free PCR hood in a pre-PCR space until the PCR Enrichment Step. Thereafter, work will be done in a post-PCR space and may be done on the bench.
- Reactions must be carried out in sterile RNase/ DNase/ DNA-free tubes.
- Once library preparation is successfully implemented at the full reaction volume, it is possible to scale to 0.5x reaction volumes by reducing the volume of all components in half.
- Prior to library preparation, prepare the mini adapter according to the [Mini Adapter Info Sheet](#).

Protocol: These protocols are an adaptation of the [Watchmaker RNA Library Prep Kit Protocol](#).

Adaptations include the addition of Qiagen FastSelect for host rRNA depletion, spike-in of ERCC synthetic RNAs as internal positive controls, and use of mini adapters instead of full-length adapters to improve ligation efficiency and reduce adapter dimer.

General notes for handling samples and reagents

- Keep all samples, buffers, and enzymes on ice unless noted
- Thaw all **buffers** on ice, then invert, vortex >5s, and spin down
- **Enzymes** and **Equinox** should be on ice before use; invert 10x
- Vortex (or pipette >10x) + spin down all master mixes
- Always centrifuge all tubes prior to opening
- Calculations in this worksheet include a 10% overage for pipetting variation

Watchmaker Illumina RNA Library Prep Worksheet

1. FastSelect + Fragmentation + Priming

Reagent	One full reaction	X reactions
		X =
	vol stock (μL)	vol stock (μL)
RNA sample (top up with water if needed)	13.5	13.5 each
25pg ERCC Spike-in (50pg/uL stock)	0.5	
Frag and Prime Buffer	10	
*QIAseq FastSelect (1:10) rRNA	1	
Total volume	25	25 each
Blue = Reagents to add to master mix		Add 11.5 uL MM to RNA sample

*FastSelect is designed for removing human rRNA, omit this reagent if not using human derived samples and only incubate at 85°C and then directly to 4°C (omit ramping down steps).

** Fragmentation is optional and dependent on input RNA quality. Refer to manufacturer protocol for guidelines. For low input or low quality samples, we recommend omitting.

Mix the master mix and final rxns by pipetting up and down.

Thermocycler (heated lid set to 105°C) - FastSelect + Fragmentation + Priming

- Pre-heat hold at 85°C
- (0-5 min at 85°C)**
- 2 min 75C
- 2 min 70C
- 2 min 65C
- 2 min 60C
- 2 min 55C
- 5 min 37C
- 5 min 25C
- Hold 4°C

2. First Strand Synthesis

Reagent	One full reaction	X reactions
		X =
	vol stock (μL)	vol stock (μL)
Fragmented & primed RNA	25	25 each
1st Strand Buffer*	9	
1st Strand Enzyme	1	
Total volume	35	35 each
Orange = Reagents to add to master mix		Add 10μL MM to frag/primed RNA

* 1st Strand Buffer is photosensitive. Protect from light.

Mix the master mix and final rxns by pipetting up and down

Thermocycler (heated lid set to 105°C) - First Strand Synthesis

- Pre-heat hold at 25°C
- 10 mins at 25°C
- 15 mins at 42°C
- 15 mins at 70°C
- Hold at 4°C

3. Second Strand Synthesis and A-Tailing

Reagent	One full reaction	X reactions
		X =
	vol stock (uL)	vol stock (uL)
First strand synthesized DNA	35	35 each
2nd Strand Buffer	14	
2nd Strand Enzyme	1	
Total volume	50	50 each
Green = Reagents to add to master mix		Add 15 uL MM to ss-cDNA

Mix the master mix and final rxns by pipetting up and down

Thermocycler (heated lid set to 80°C) - Second Strand Synthesis and A-Tailing

- Pre-heat hold at 42°C
- 5 mins at 42°C
- 10 mins at 62°C
- Hold at 4°C

During incubation, prepare mini adapter

4. Adapter Ligation

Ensure adapters are at appropriate concentration. Dilute adapters prior to making master mix if needed.

Add adapter separately after ligation buffer and ligation enzyme to avoid adapter dimers.

Reagent	One full reaction	X reactions
		X =
	vol stock (uL)	vol stock (uL)
ds-cDNA	50	50 each
Ligation Buffer*	40	
Ligation Enzyme	5	
Diluted adapter	5	-
Total volume	100	100 each
Pink = Reagents to add to master mix		Add 45 uL mix to ds-cDNA
		Add 5uL diluted adapter to mix

* Vortex the thawed Ligation Buffer for 20 sec to fully homogenize, then centrifuge to collect at the bottom of the tube and place it on ice.

Thermocycler (heated lid off) - Adapter Ligation

- Pre-heat hold at 20°C
- 15 mins at 20°C
- Hold at 4°C

5. SPRI Bead Clean-up and Size Selection

SPRI bead ratio:	0.9
Sample volume: (<i>adapter-ligated cDNA</i>)	100.00
SPRI to add:	90.00
Elution volume:	22
Aspiration volume:	20

SPRI clean-up protocol:

*allow beads to sit in RT for 30 mins prior. Ensure you have enough 80% ethanol.

1. Add room temperature beads to rxn. Mix well by pipetting up and down 10 times.
2. Incubate for 5 mins at room temperature.
3. Place samples on magnetic rack, and incubate for 5 mins on the rack.
4. Remove supernatant
5. Add 200uL of 80% EtOH to samples while on the magnetic rack. Do not disturb beads.
6. Incubate at room temperature for 30s and remove the supernatant.
7. Repeat EtOH wash step once for a total of 2 washes.
8. Remove any excess EtOH with a p10 or p20 pipette.
9. Air dry the beads for 2-5 mins while on the magnetic rack.

Watch beads carefully throughout this incubation to avoid overdrying. Smaller bead pellets dry faster. Trust your eyes more than the timer.

10. Remove tube from magnetic rack. Add desired elution volume of nuclease-free water.
11. Pipette to mix well and incubate for 2 mins at room temperature off the magnetic rack.
12. Pulse spin tubes and place on magnetic rack until solution is clear, ~ 5 mins.
13. Aspirate supernatant and transfer to a clean nuclease free PCR tube.

**Checkpoint: Samples can be stored frozen at -20°C and library prep resumed the next day.

6. Barcoding PCR

Thaw Equinox master mix on ice, then vortex and spin

Reagent	One reaction	X reactions
		X =
	vol stock (ul)	vol stock (ul)
Purified, adapter-ligated ds-cDNA	7.5*	7.5 each
Equinox Amplification Master Mix (2x)	12.5	12.5 each
5uM each i5/i7 barcoded primers (NEB index primer/TruSeq/or similar)	5 (*premixed)	5 unique for each sample
Total volume	25	25 each

* Note: only a fraction of the sample volume is input to the Barcoding PCR

Record adapter barcode assignments

Thermocycler (lid at 105°C)	Cycles
98 °C for 45s	1
98 °C for 15s	7-15**
55 °C for 30s	
72 °C for 30s	
72°C for 1 min	1
Hold at 10°C	

** # of PCR cycles is dependent on RNA input and adapters used.
Refer to manufacturer protocol below for guidelines

Table A4. Recommended PCR cycle numbers by RNA input amount into library prep

RNA input into Library Preparation (ng)	PCR cycles to generate 10 – 50 nM library	
	Full-length adapters	Stubby adapters
100	8	7 ¹
50	8 – 9	
25	9 – 10	8
10	10 – 11	8 – 9
5	11 – 12	9 – 10
1	14 – 15	10 – 11
<1	18 – 19	14 – 15

¹A minimum of 6 amplification cycles is recommended to ensure transcript strand origin information is maintained.

7. SPRI Bead Clean-up and Size Selection

SPRI bead ratio:	0.9
Sample volume: (<i>Barcoded Library</i>)	25.00
SPRI to add:	22.50
Elution volume:	22
Aspiration volume:	20

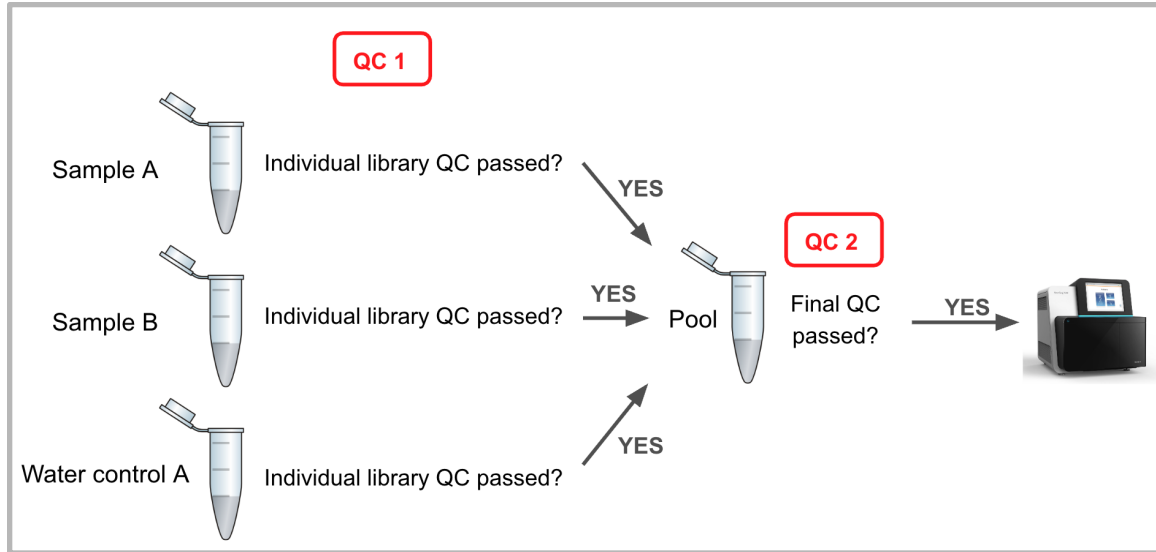
*See Step 5 for SPRI clean-up protocol

**Checkpoint: Samples can be stored frozen at -20 °C and library prep resumed the next day.

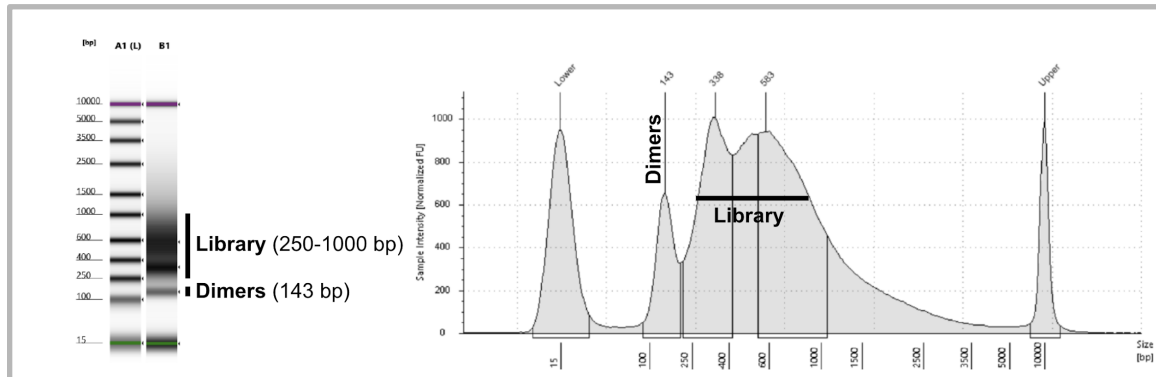
WL- Protocol 3: Illumina RNA Library QC & Pooling

Goal

Determine the quality of the finished libraries, pool libraries and quantify the final pooled libraries



Adapter dimers need to be removed before the library can be sequenced.



Workflow

- I. **Assess quantity** with High sensitivity DNA Qubit
- II. **Assess size** with Tapestation or by Diagnostic PCR + Gel Electrophoresis
- III. **Interpret QC results**
Secondary amplification (as needed) - *see appendix*
- IV. **Pool Libraries**

Considerations and Protocols

Remove all adapter dimers by SPRI bead size selection prior to doing a final quantification of the pooled library. qPCR and/or Qubit quantification of the final library are the gold standard for loading onto Illumina sequencers.

I. Quantify libraries with Qubit

This protocol is for the Qubit 1X dsDNA HS Assay Kit. If using a different kit, prepare working solution in advance.

1. Set up the required number of PCR tubes for 2 standards and all samples. *Note: Label the tube lids. Do not label the side of the tube as this could interfere with the sample read.*
2. Add 10 μ L of each Qubit standard to the appropriate tube.
3. Add 1 μ L of library to each tube.
4. Add working solution to standards and samples.
 - Standard tubes: 190 μ L of working solution
 - Sample tubes: 199 μ L of working solution for 1 μ L of sample.
 - Mix by vortexing 2-3 seconds, then spin down.
5. Incubate for at least 3 minutes and then proceed to read on the Qubit machine.
 - *Water controls should have substantially less or unquantifiable concentrations than the rest of the samples.*

II. Assess size and presence of adapter dimers in libraries by Tapestation

1. Launch the Controller Software
2. Load D5000 ScreenTape and loading tips into the Tapestation
3. Vortex and spin down ladder and reagents
4. Prepare ladder in the first tube of the strip by mixing 1ul D5000 ladder and 10ul D5000 buffer
5. Prepare your samples in the tubes following the ladder by mixing 1ul of sample library and 10ul D5000 buffer
6. Spin down, then vortex for 1 min
7. Spin down samples and load into the Tapestation
8. Select the required samples on the Controller Software
9. Click Start and specify a filename with which to save your results
10. When the run is complete, create the following regions under Region Settings:
 - 100-200bp, 200-1000bp
11. Export the results. File -> Create Report

II. Assess size and presence of adapter dimers in libraries by Diagnostic PCR

This is an alternative to the Tapestation option

Perform a standard PCR reaction with 1 uL of the sample for 30 cycles using the Illumina universal primers:

P7: CAAGCAGAAGACGGCATACGAGAT

P5: AATGATACGGCGACCACCGAGATCTACAC

	0.5x	__x
2x PCR master mix	12.5 uL	
P5 + P7 primer mix (5µM each)	1 uL	
Library sample	1 uL	
Water	10.5 uL	
Total volume	25	

Cycling

conditions for diagnostic PCR:

98C 1 min
98C* 10s
60C* 15s x 30 cycles
72C* 75s
72C 5 min
4C hold

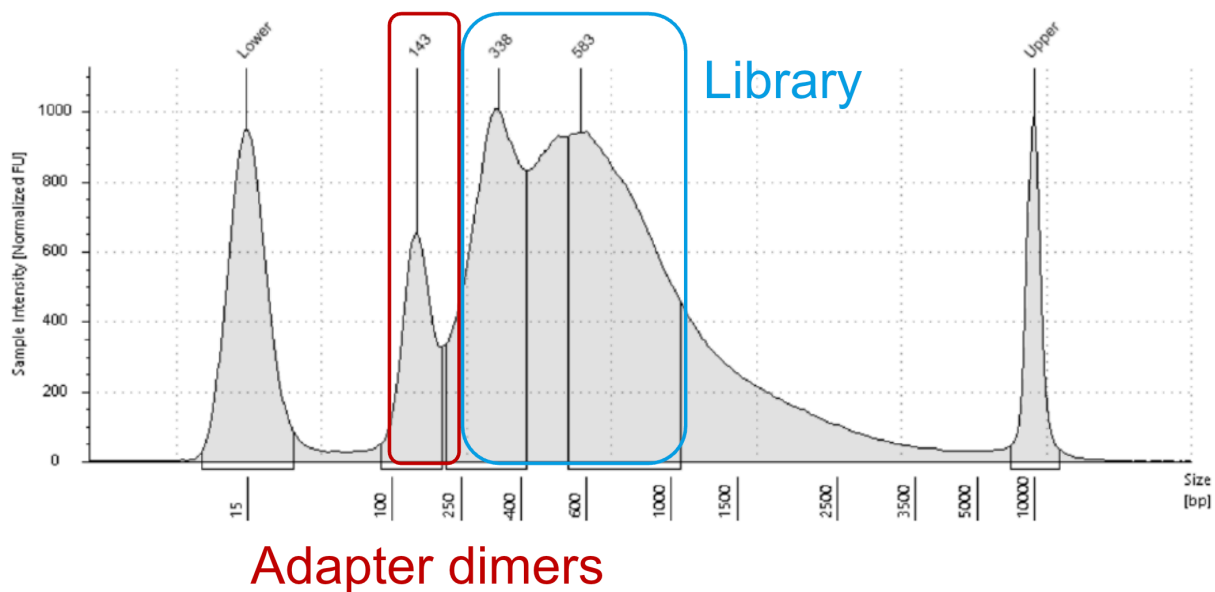
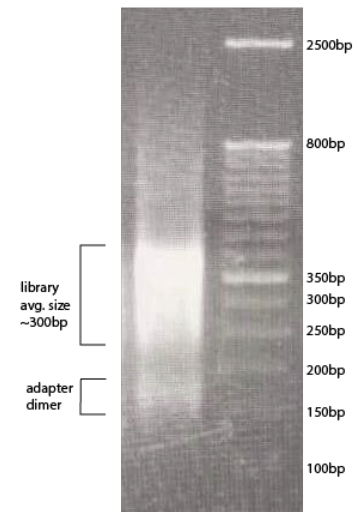
- * Thermocycling conditions will vary depending on the polymerase used. Be sure to refer to the PCR kit recommendations and calculate T_m for the P5, P7 primers.

When the PCR is complete, run 10uL of the PCR reaction and 10uL of a 1:5 dilution of the PCR reaction on a 2% agarose gel to assess the presence or absence of the adapter dimer.

III. Interpreting the QC results

Your libraries should have an average size of ~300-500 bp, have at least 4nM, and not have any remaining adapter dimers. If this is not the case, refer to the decision tree below.

- Adapter dimers show up ~146bp. If adapter dimers are present, clean up the original library (not the diagnostic PCR) using a 0.8x SPRI clean-up. If >10% of the sample is adapter dimer, it will typically take 2 or even 3 clean-ups before all the adapter dimers are removed.
- If one or more samples fail by Qubit due to very low concentration, you can attempt to perform secondary library amplification (see appendix). SPRI clean the amplified library with a 0.8x SPRI ratio and repeat QC.



IV. Pooling Libraries

Pool the samples and water controls. Samples should be pooled at equivalent amounts and water controls to 1/5th or 1/10th the amount of a real sample. For example, pool 10ng from all the samples and 1-2ng of the water controls.



<u>Qubit concentration</u>		<u>Pool</u>			
Sample 1:	10 ng/ul	->	4.0 µl	->	40 ng
Sample 2:	5 ng/ul	->	8.0 µl	->	40 ng
Sample 3:	8 ng/ul	->	5.0 µl	->	40 ng
Sample 4:	15 ng/ul	->	2.6 µl	->	40 ng
H ₂ O control:	1 ng/ul	->	8.0 µl	->	8 ng
Total:			27.6 µl		168 ng

End goal is >10 ul of 4 nM pooled library

$$\text{Concentration in nM} = \frac{\text{concentration in ng/}\mu\text{l}}{660\text{g/mol} \times \text{average library size in bp}} \times 10^6$$

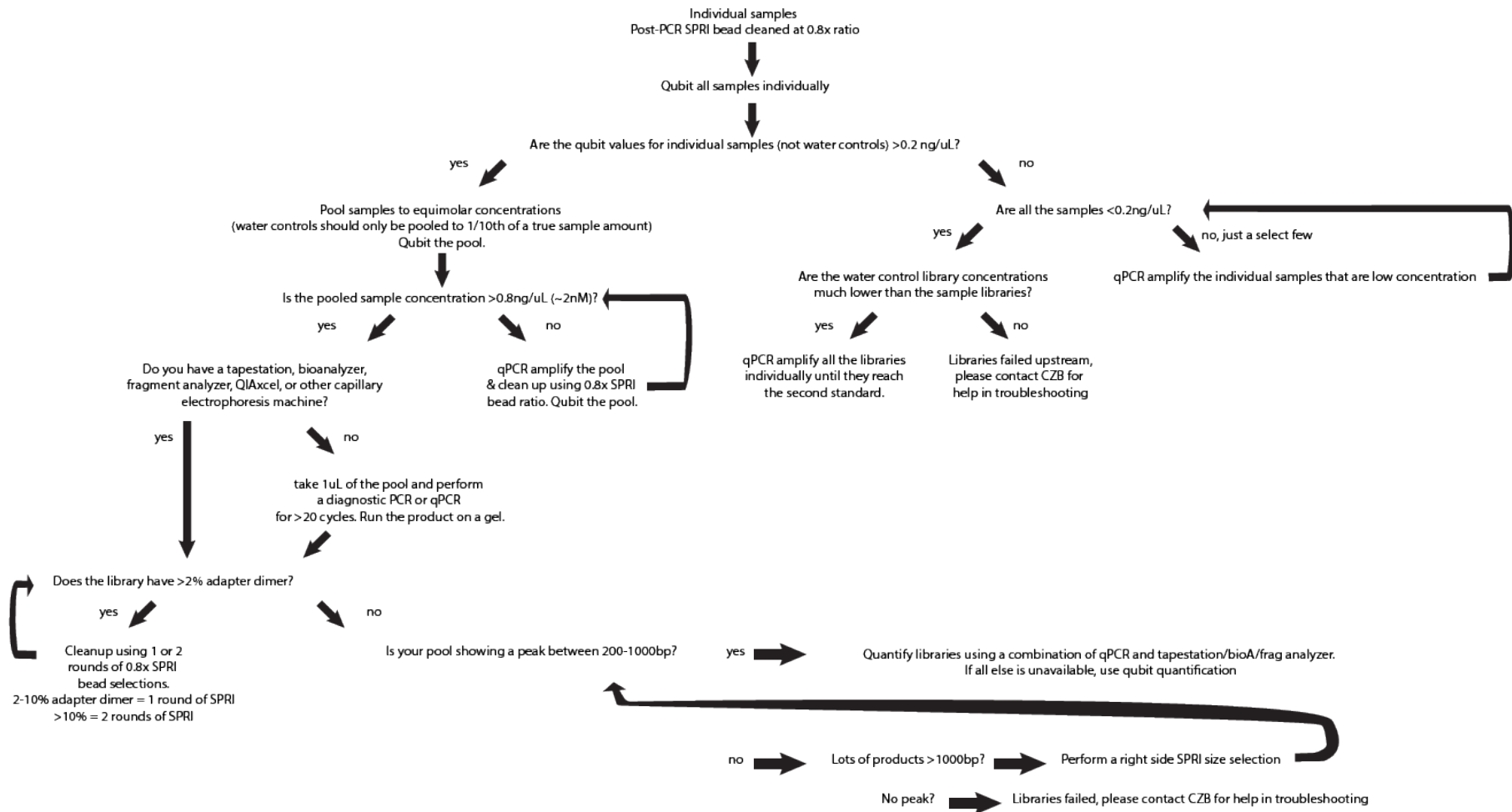
↑
Capillary or gel electrophoresis

Example: if concentration is 6.1ng/ul and average library size is 550 bp -> concentration is 16.8 nM.

Pooling guidelines

- Do not pool more than half of your library volume (e.g. if you have 15ul of library, do not pool more than 7ul maximum)
- Pool waters/negative controls less mass than samples (1/5th or 1/10th)
- Do not pipette less than 2ul. Adjust input/calculations or dilute samples if pipetting volume is <2ul

V. QC Decision Tree



WL- Protocol 4: Loading the Sequencer

Goal

Sequence library pool on an Illumina instrument

Before starting

- Always refer to the manufacturer-published protocol for complete information:
- If relevant, prepare aliquots of PhiX, stored at -20C
- Prepare library pool to starting concentration
- Choose loading concentration
 - *This will depend on the instrument / sequencing kit*
- Determine desired % PhiX spike-in for the run
 - *Note: we typically use 1% for mNGS because they are diverse libraries. A higher amount of PhiX is recommended for low diversity libraries*
- Prepare calculations for denature and dilute protocol, if relevant

Workflow

- I. **Prepare Sample Sheet** (all platforms)
- II. **Denature** (MiniSeq, MiSeq only) **and Dilute Library Pool**
- III. **Prepare and Load the Cartridge for Sequencing** (see Illumina documentation)
- IV. **Set Up a Sequencing Run**

I. Prepare Sample Sheet

What is a sample sheet?

A sample sheet is a comma-separated value (*.csv) file format used by Illumina instruments, platforms, and analysis pipelines to store settings and data for sequencing and analysis. It specifies the **names of samples** in the sequencing run along with their **corresponding barcode sequences**. This is used by Illumina instruments and analysis pipelines for **demultiplexing** sequencing data.

It is possible to manually enter sample names and barcodes during the sequencing setup, but this can be error prone, so using a sample sheet is preferred.

A sample sheet is not always necessary to *perform* sequencing, it is generally only needed during demultiplexing which is done after sequencing has been completed.

We recommend consulting Illumina documentation for the most up to date information and/or modifying a template sample sheet that you have successfully used in a previous run on the same instrument.

II. Denature and Dilute Library Pool

Consult Illumina documentation

III. Prepare and Load the Cartridge for Sequencing

Consult Illumina documentation

IV. Quick Guide – Set up a Sequencing Run:

There are four important parameters to communicate to the sequencing instrument:

1. Basespace

- For run monitoring and storage

**Make sure you have an SOP for routinely cleaning Basespace and sequencer local storage*

2. Cycle parameters

- (read1 | index1 | index2 | read2)
- (bp): 146 | 12 | 12 | 146

**Read and index lengths may vary based on library prep reagents and sequencing kit used*

3. Sample sheet

- Indexes used for each sample

**Make sure i5 is in the correct orientation. See [i5 Index Orientation Table](#)*

4. Workflow

- GenerateFASTQ workflow to queue demux.

This information is often included in the sample sheet

Activity: Basespace or SAV Run Review

Hint: use the slides from the Basespace run QC session as a guide

1. Look at **Clusters Pass Filter and Average Q30**. How did the run look, overall? Was this a good / okay / bad sequencing run?

2. **How much data** did you get back (reads)? How does this compare to what you expected?

Hint: Google your sequencing kit to find expected output

3. Which metric(s) will you look at to check your **loading concentration**?

Hint: Look at % aligned

Hint: Does the sequencer used use a patterned or non-patterned flow cell?

Non-patterned: look at density; patterned look at % occupied

- a. Do you think enough library pool was loaded?
 - b. Next time would you load more, less, or the same amount?
4. What is another way to look at **clustering**? How did it look for this run? Over-, under-, just right?
Hint: Thumbnail images (available for some sequencers)

5. How did the **library composition** look? What metric/chart will you look at to evaluate this? Do you see evidence of **adapter dimer**? Do you have short fragments?

Hint: % Base chart

6. How was the **index distribution** and how does this compare to the expected based on sample pooling? *Did your group's samples receive equal/expected percentage of reads?*

7. If you got 100% of your reads labeled as "Undetermined", what issue do you suspect? How would you correct this?

8. If you do not have the sequencer connected to the internet or Basespace, what tool can you use to look at your run metrics?

Hint: Sequencing Analysis Viewer (SAV)

9. Based on your answers above, how would you improve your next sequencing run?

MODULE 2: **ANALYSIS and INTERPRETATION**

*Analysis and Interpretation of Sequencing
Data Utilizing CZ ID and Other Bioinformatics
Tools*

CZ ID Training: Reference Tables

QC Metrics Table

QC Metric	Definition
Passed Filters	The percentage of reads remaining after host filtering, quality filtering and subsampling, compared to the initial reads input. Tells you how many reads were used for final analysis.
Passed QC	The percentage of reads that passed the quality filtering thresholds imposed by fastp. Gives you an idea of the sequencing quality.
Total Reads*	The total number of reads uploaded.
DCR*	Duplicate Compression Ratio; The ratio of sequences present prior to running CZID dedup (duplicate removal) vs after duplicate removal.
ERCC reads*	The total number of reads aligning to ERCC (External RNA Controls Consortium) sequences. The number of ERCC reads will be proportionally higher in samples with relatively low input RNA as compared to those with high amounts of input RNA.
SubSampled Fraction*	After host filtration and QC, the remaining reads are subsampled to 1 million fragments (2 million paired reads). This field indicates the ratio of subsampled reads to total reads passing host filtration and QC steps.

** Fields are not default in the samples table and need to be added by clicking on the column selector*

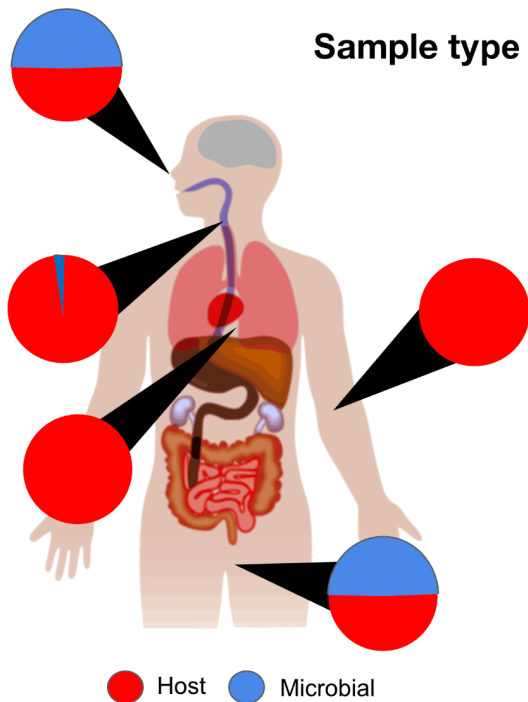
Sample Metrics Table

Report Metric	Definition
Score	Score is only calculated if a background model is applied. It is CZ ID's heuristic for ranking microbial hits. The score is intended to combine the following aspects of the evidence for a hit: (a) species-level information, (b) genus-level information, (c) information about relative abundance within the sample, (d) information about abundance relative to the chosen background controls. The score is calculated as follows: $((\text{abs}(\text{genus NT Z}) * \text{species NT Z} * \text{species NT rPM}) + (\text{abs}(\text{genus NR Z}) * \text{species NR Z} * \text{species NR rPM}))$
Z	Z-score statistic is only computed if a background model is applied. It is used for evaluating the prevalence of microbes in your sample as compared to background controls. The Z-score is computed based on the specified background model.
rPM	Number of reads aligning to the taxon in the NCBI NT/NR database, per million reads sequenced
r	Number of reads aligning to the taxon in the NCBI NT/NR database

contig	Number of assembled contigs aligning to the taxon in the NCBI NT/NR database
contig r	Total number of reads aligning to all assembled contigs for this taxon
%id	Average percent-identity of alignments to NCBI NT/NR
L	Average length of the local alignment for all contigs and reads assigned to this taxon
E value	Average Expect value (e-value) of alignments to NCBI NT/NR. The Expect value (e-value) is a parameter that describes the number of hits one can "expect" to see by chance when searching a database of a particular size. The closer to 0 the better.

Pipeline Table

Pipeline Step	Function	If Many Reads Lost During this Step...
fastp	Adapter Trimming	Many short reads, possible adapter dimers, possible degradation, possible over-fragmentation.
	Low quality read removal	Many reads of low-quality (Phred scores, GC content, over-clustering).
	Low complexity read removal	Reads are relatively low-complexity - containing homopolymer nucleotides or simple sequence repeats
HISAT2, Bowtie 2	Host Subtraction	Large proportion of reads mapped to the host reference genome
HISAT2, Bowtie 2	Remove human reads regardless of host	Large proportion of reads mapped to human reference genome
CZID-dedup (*Available through "reads remaining")	Remove duplicates	Many duplicate reads, possible over-amplification during library preparation.



Sample type matters when evaluating QC metrics

Sample Type	Host Signal	Microbial Signal
CSF	> 99%	<< 5% with infection
Tracheal Aspirate	95%	5%
Nasal or oral	50%	50%
Whole blood	> 99%	<< 1% (with infection)
Rectal or Vaginal	50%	50%

QC metrics affected by sample type

Sample Type	Typical Distribution ● Host ● Microbial ● ERCC	QC Trend
Stool, NP Swab, Environmental Samples, Sputum, Urine (Higher diversity & DNA/cDNA input)		- Higher microbial reads - Lower DCR
Serum, CSF, Plasma, Negative Controls (Lower diversity & DNA/cDNA input)		- Lower microbial reads - Higher DCR (PCR artifact) - Higher background contamination

CZ ID Training: Worksheets

Goal For Your Project

Why is your team using metagenomics and CZ ID? What problem(s) are you trying to solve with mNGS and how does it relate to your broader research?

QC Worksheet

QC Step 1: Identify samples with low total reads

The total number of reads uploaded (after input validation).

Metric	Criterion	How to Interpret	Sample Name(s)	Next Steps/Notes
Total Reads	Outliers	You should make note of outliers and high variability in Total Reads. This can indicate uneven pooling in the library preparation steps. You may choose to resequence low-output samples.		

QC Step 2: Identify samples with low-quality reads (low passed QC)

The percentage of reads that passed the quality filtering thresholds imposed by fastp. Gives you an idea of the sequencing quality.

Metric	Criterion	How to Interpret	Sample Name(s)	Next Steps/Notes
Passed QC	>50%	You can be relatively confident in the quality of the sequencing data.		
	<10%	Samples with low % Passed QC likely contained many short sequences that were associated with adapter dimers or sequences that obtained low-quality scores during base-calling. If the Passed Filters value is below 20%, you should work to understand why reads were filtered out (see next section) and use this information to improve library preparation steps. Interpret results from these samples with caution.		
	*Outliers	Take note of any values above that stand apart from the others because of very high or very low Passed QC values. These should be noted for their unusually high or low sequencing quality and looked into more closely.		

QC Step 3: Identify samples with high number of duplicates

The ratio of sequences present prior to running cd-hit-dup (duplicate removal) vs after duplicate removal.

Metric	Criterion	How to Interpret	Sample Name(s)	Next Steps/Notes
DCR	>3	There were a lot of duplicate reads in your sample, indicating lower library complexity - this can be due to PCR amplification and jackpotting.		What is the Sample Type for each of the samples? • Samples with a known microbiome (ie Stool, NP Swab, Environmental Samples, Sputum) tend to have a lower DCR due to the complexity of the samples and abundance of total nucleic acid. • Sterile site fluids (Serum, CSF, Plasma) may have a higher DCR due to over-amplification of low biomass samples.

QC Step 4: Identify samples with low number of reads passing CZ ID filters

The percentage of reads remaining after host filtering, quality filtering and subsampling, compared to the initial reads input. Tells you how many reads were used for final analysis.

Metric	Criterion	How to Interpret	Sample Name(s)	Next Steps/Notes
Passed Filters	>10%	A higher Passed Filters value is generally positive as it indicates that you have a greater amount of reads to analyze for non-host taxa.		What is the Sample Type for each of the samples? • Samples with a known microbiome (ie Stool, NP Swab, Environmental Samples, Sputum) tend to have a higher number of non-host reads due to greater microbial content in the samples. • Sterile site fluids (Serum, CSF, Plasma) tend to have fewer non-host reads and greater numbers of host or ERCC reads. • Whole Blood tends to have a distribution of non-host reads in between these two extremes due to the high abundance of host cells.
	<5%	It is not uncommon to see Passed Filters values below 1% for samples with high abundance of host reads.		
	*Outliers	Take note of any values that stand apart from the others because of very high or very low Passed Filters values. These should be noted for their unusually high or low amount of non-host data and looked into more closely.		

QC Step 5: Evaluate ERCC Spike-Ins

The total number of reads aligning to [ERCC](#) (External RNA Controls Consortium) sequences. The number of ERCC reads will be proportionally higher in samples with relatively low input RNA as compared to those with high amounts of input RNA.

Metric	Criterion	How to Interpret	Sample Name(s)	Next Steps/Notes
ERCC reads	Outliers / unexpected proportion based on sample type	What is the Sample Type for each of the samples? - Samples with a known microbiome (ie Stool, NP Swab, Environmental Samples, Sputum) tend to have higher microbial/host content and, therefore, low ERCC reads. - Sterile site fluids (Serum, CSF, Plasma) are expected to have higher ERCC reads due to low biomass samples.		
ERCC plots	Poor correlation of spike-in vs detected read values			

Sample QC Summary Table

Identify samples with poor QC metric(s) to flag for analysis interpretation

Using the reads lost chart and QC Steps 1-4 above, fill out the following for low quality samples:

Sample Name	Step(s) where lost many reads	ERCC Spike In Counts pass checks?	Conclusions
		☐	
		☐	
		☐	
		☐	
		☐	

Sample Report

Think about the following questions when looking at your sample report:

1. What is the goal of your study?
2. Are there any pathogens present that could be causing disease in your set of samples?
 - a. Is it a real hit? Why do you think it is?
3. What are some of the other microbes?
 - a. Why do you think they are there?
4. Did you find any false positives?
 - a. How did you determine this?

Sample Summary Table (#1)

Name of sample

Taxon Name	Important Metrics (i.e. high rPM, known pathogen, number of contigs, length)	Pathogenic?	Does it answer your research question?	Coverage Visualization Notes	BLAST Notes	Why do you think this microbe is present?	Is this a real hit?
		☐					
		☐					
		☐					
		☐					

Sample Summary Table (#2)

Name of sample

Taxon Name	Important Metrics (i.e. high rPM, known pathogen, number of contigs, length)	Pathogenic?	Does it answer your research question?	Coverage Visualization Notes	BLAST Notes	Why do you think this microbe is present?	Is this a real hit?
		☐					
		☐					
		☐					
		☐					

Sample Summary Table (#3)

Name of sample

Taxon Name	Important Metrics (i.e. high rPM, known pathogen, number of contigs, length)	Pathogenic?	Does it answer your research question?	Coverage Visualization Notes	BLAST Notes	Why do you think this microbe is present?	Is this a real hit?
		☐					
		☐					
		☐					
		☐					

Sample Summary Table (#4)

Name of sample

Taxon Name	Important Metrics (i.e. high rPM, known pathogen, number of contigs, length)	Pathogenic?	Does it answer your research question?	Coverage Visualization Notes	BLAST Notes	Why do you think this microbe is present?	Is this a real hit?
		☐					
		☐					
		☐					
		☐					

Heatmap Notes

Suggested starting threshold filters: NT rPM ≥ 10 , NR rPM ≥ 1 , NT L > 50

	Initial Heatmap (no filters)	With Background Model + threshold filters	Background model + threshold filters + category filter
Are there any organisms present in the water samples? If so, are they present in any other samples?			
Which organisms appear consistently across all samples, but not in water control?			
Are there any pathogens present that you want to look into?			

If you have metadata, are there microbes present in one set of samples that aren't in another (i.e. cases vs controls)?			
--	--	--	--

Space for other notes on the Heatmap

Appendix: DNase I treatment protocol

Many brands of DNase I are sufficient. Below are protocols for two options, but we recommend following the manufacturer protocol for the type that is available to you.

DNase treatment - Zymo

Reagent	87.5 µL sample	20 µL sample	X 20µL rxns
			X = 10
	vol stock (µL)	vol stock (µL)	vol stock (µL)
TNA sample	40	20	20 each
DNA Digestion Buffer (Zymo)	5	2.50	25.00
DNase I stock (1U/ul) (Zymo)	5	2.50	25.00
Master Mix volume	10	5.00	50.00
Total volume	50	25.00	25 each
Green = Reagents to add to master mix			Add 5 µL mix to TNA sample

Incubate for 15min at RT
Complete a SPRI clean-up with a 2x SPRI bead ratio

SPRI Bead Clean-up - Zymo DNase

SPRI bead ratio:	2	
Sample volume (per half rxn):	25	µL
SPRI to add (per half rxn):	50	µL
Elution volume (per half rxn):	12	µL
Aspiration volume (per half rxn):	10	µL

or

DNase treatment - Qiagen

Reagent	87.5 µL sample	20 µL sample	X 20µL rxns
			X = 10
	vol stock (µL)	vol stock (µL)	vol stock (µL)
TNA sample	87.5	20	20 each
RDD Buffer (Qiagen)	10	2.29	22.86
DNase I (Qiagen)	2.5	0.57	5.71
Master Mix volume	12.5	2.86	28.57
Total volume	100	22.86	22.86 each
Blue = Reagents to add to master mix			Add 2.86 µL mix to TNA sample

Incubate for 30min at RT
Complete a SPRI clean-up with a 2x SPRI bead ratio

SPRI Bead Clean-up - Qiagen DNase

SPRI bead ratio:	2	
Sample volume (per half rxn):	22.86	µL
SPRI to add (per half rxn):	45.71	µL
Elution volume (per half rxn):	12	µL
Aspiration volume (per half rxn):	10	µL

Appendix: Secondary Amplification protocol

[Appendix Hb](#). Library amplification using KAPA HiFi HotStart Library Amp Kit (KK2611/KK2612)

This protocol is based on the [KAPA HiFi HotStart Library Amp Kit](#) (KK2611), which does not include fluorescent dyes and standards. Given this, the amplification can be performed on a regular PCR machine. However, without the fluorescent standards, the number of amplification cycles needs to be pre-determined.

- Perform this PCR amplification step if after cleaning out the digested fragments and adapter dimers your library concentration is too low for loading onto a sequencer.
- Quantify the concentration of the initial library using Qubit.
- Thaw the reagents and P5/P7 universal Illumina primers
P7: CAAGCAGAAGACGGCATACGAGAT
P5: AATGATACGGCGACCACCGAGATCTACAC
- For each library to be amplified, setup the amplification reaction as follows:

Component	Volume (µL)
KAPA Master Mix (2x)	25
P5/P7 primer mix (5µM)	2
DNA (bring up to volume with H ₂ O if necessary)	23
Total Volume	50

- Determine the number of cycles of PCR needed based on the table below. This table is from the [KAPA amplification kit manual](#). **Note: from our experience, fewer cycles may be needed than is indicated. For low inputs (<10 ng), amplifying for two cycles fewer than the lower range of what is recommended in the table seems sufficient*.**

Table 1: Recommended library amplification cycles to generate 100 ng or 1 µg of amplified library, constructed from different amounts of input DNA with a KAPA HyperPrep or KAPA HyperPlus Kit

Input DNA (into library construction)	Number of cycles required to generate 100 ng – 1 µg			
	KAPA Hyper Prep		KAPA HyperPlus	
	100 ng library	1 µg library	100 ng library	1 µg library
1 µg	0	1 – 2	0	0 – 1
500 ng	0	2 – 4	0	2 – 3
250 ng	1 – 2	4 – 6	0 – 1	3 – 5
100 ng	2 – 3	6 – 7	0 – 2	5 – 6
50 ng	3 – 5	7 – 8	3 – 5	7 – 8
25 ng	5 – 7	8 – 10	5 – 7	8 – 10
10 ng	7 – 9	11 – 13	7 – 9	11 – 13
5 ng	9 – 11	13 – 14	9 – 11	13 – 14
2.5 ng	11 – 13	14 – 16	11 – 13	14 – 16
1 ng	13 – 15	17 – 19	13 – 15	17 – 19

**For example, if the Qubit value for a library is 0.2 ng/µL, the total input amount is 0.2*23 = 4.6 ng. This is closest to 5 ng, and the lower range for this input amount is 9-11 cycles. Subtracting two, 7 cycles of amplification are likely sufficient.*

- Use the following cycling conditions for PCR:

Temperature	Time	Cycles
98°C	45 sec	1
98°C	15 sec	X*
63°C	30 sec	
72°C	1m 45 sec	
72°C	20 sec	

*X is determined based on input, as indicated above.

- After amplification, perform a 0.8x or 0.9x SPRI clean-up and then quantify and visualize the size distribution to assess quality and quantity of the library.

TruSeq-compatible mini adapters

Background and introduction

Ordering

Storage and Handling

Preparation

Mini adapter CZB pilot

References

Background and introduction

Adapter dimer is an unwanted byproduct of ligation-based library preparation, especially when working with low input samples. Size-specific SPRI bead cleanups are used to remove the dimers which are typically smaller than the target library size (<200bp). However, dimer can persist which interferes with sequencing output and quality.

To address this issue of adapter dimer, we share new TruSeq-compatible mini adapters. These adapters can swap into the current [CZB mNGS library preparation protocol](#) (and others) with minor modifications and are compatible with TruSeq-based indexing primers.

The adapters are shorter than the typical NEB/Illumina adapters so they can be used at higher concentrations and still be removed efficiently during the SPRI bead clean after adapter ligation. Mini adapter dimer is ~50bp whereas traditional adapter dimer is ~150bp.

Based on our pilot, we recommend piloting these mini adapters and implementing in place of the NEB adapters.

Ordering

The mini adapters may be ordered from IDT or another oligo manufacturer.

Product	Purification	Name	Sequence	Bases
100 nmole DNA Oligo	PAGE Purification	TruSeq mini adapter R1	CGA CGC TCT TCC GAT C*T	17
100 nmole DNA Oligo	PAGE Purification	TruSeq mini adapter R2	/5Phos/GAT CGG AAG AGC ACA CG*T	18

Storage and Handling

The mini adapters should be stored at -20C and kept on ice when at the bench, especially after annealing is completed. The mini adapters have a 5' phosphate which can be fragile when handled at room temperature.

Preparation of mini adapters

1. [Reconstitute lyophilized adapters](#) received from manufacturer in NEB adapter dilution buffer to 100uM in original tube. **Example:**
 - TS mini adapter **R1**: 20nmol
 - Add 200ul of NEB adapter dilution buffer for 100uM final
 - TS mini adapter **R2**: 15nmol
 - Add 150ul of NEB adapter dilution buffer for 100uM final

2. Make a 15uM mix of the two adapters

In a fresh strip tube, combine the following components:

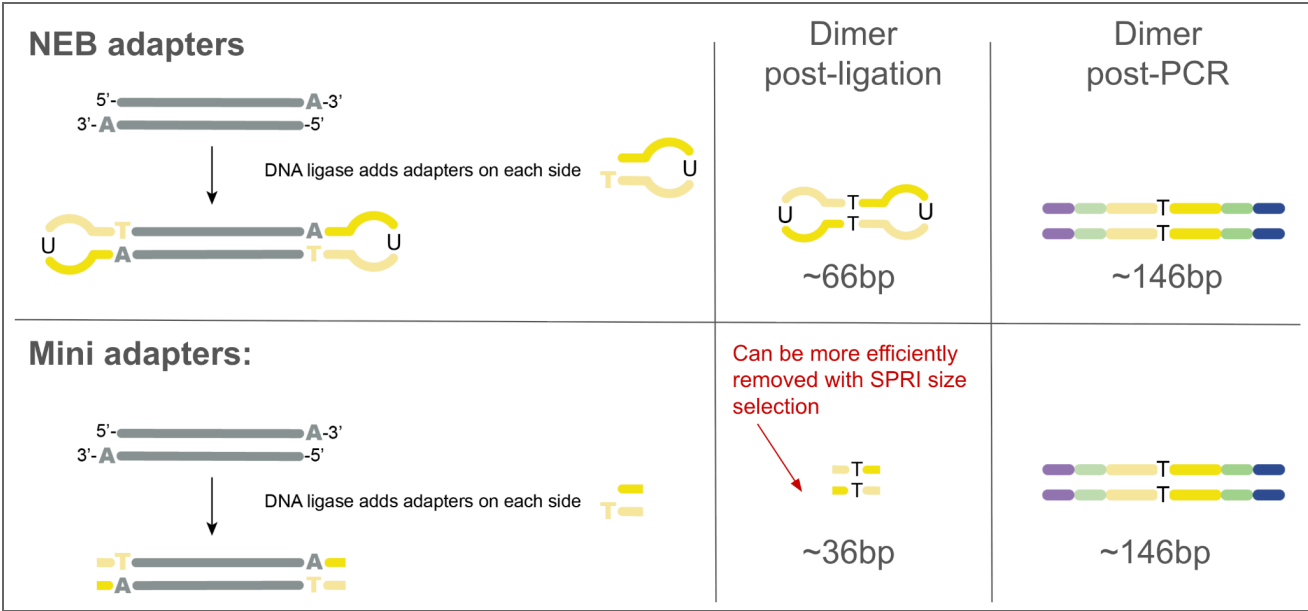
Component	ul to add
TS mini adapter R1 (100uM)	7.5
TS mini adapter R2 (100uM)	7.5
NEB adapter dilution buffer	35
Total volume of 15uM mini adapter mix	50

3. Anneal adapters
 - Incubate the 15uM mix in the thermocycler at 70C for 5 minutes, then end the program and let the thermocycler block naturally cool to room temperature. Do not snap cool.
4. Dilute annealed adapters to 3uM working solution. Keep cold.

Component	ul to add
15uM annealed mini adapters	50
Cold NEB adapter dilution buffer	200
Total volume of 3uM annealed mini adapter mix	250

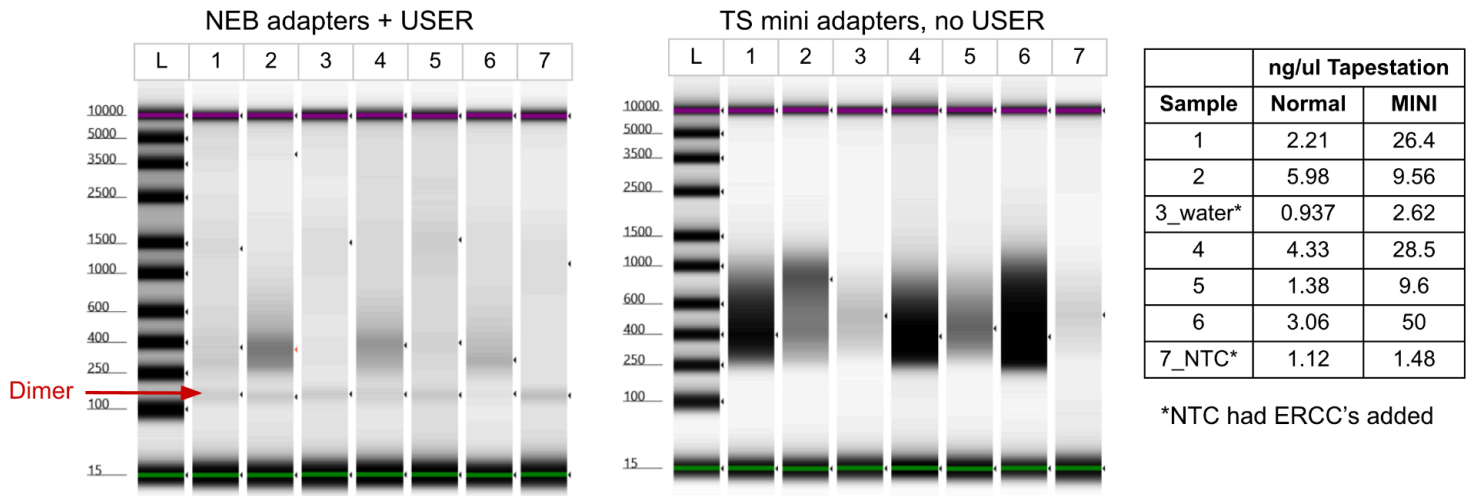
5. Keeping the tubes and adapter cold, make 25ul aliquots of ready-to-use 3uM annealed mini adapter mix. Store at -20C.

Diagram of adapter dimer in library prep



TS mini adapter CZB pilot

Samples: nasal swab RNA
Libraries: prepped according to standard [CZB Rapid Response NEB RNA mNGS protocol](#)
Conditions: 1. NEB adapters 1:200 + USER; 2. TS mini adapters, no USER
Shown below: individual libraries cleaned once 0.8x after Q5 index PCR



Conclusion: less dimer and higher concentration libraries are observed in libraries prepared with mini adapters compared to NEB adapters

